Laboratory Work 4

Graded Practice:

Arithmetic, Types and `if/else`

21 problems + 4 bonus from LeetCode · score out of 100 · checked by the TA at your desk

Week: 3 — second session (graded)
Duration: 2 hours 15 minutes
Format: individual problem solving in CLion
Covers: Lab 3 (types, `cin`/`cout`, `/`, `%`) and the Week 3 lecture (operators, `if/else`)

Name:

Group:

Date:

How this lab works

1. Solve the problems in CLion, one `.cpp` file per problem. Every program reads its input with `cin` and prints its answer with `cout`.
2. **When a problem works, raise your hand.** The TA comes to your desk, reads your code, runs a few inputs and writes your points in the box under the problem. Nothing is uploaded.
3. You may ask the TA for help — this is practice, not an exam. The TA may ask you a short question about your code; be ready to explain *why* it works.
4. There are **170 points** on offer but your score is **capped at 100** — you do not have to solve everything. Choose problems you can do; Part A is easier than Part B. A problem with no attempt earns 0 points.

All problems are from `leetcode.com`, restated for a program that reads from the keyboard. On LeetCode itself the answer is written as a *function* inside `class Solution`; you will learn functions in Week 6 and can then submit these problems there for extra practice.

Quick reference

Program skeleton

```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b;
    cin >> a >> b;          // input
    int sum = a + b;        // process
    cout << sum << endl;    // output
    return 0;
}
```

Types `int` (up to about 2.1×10^9), `long long` (up to about 9.2×10^{18}), `double`, `char`, `bool`, `string`.

Integer arithmetic `17 / 5` is `3`, `17 % 5` is `2`, `n % 10` is the last digit, `n / 10` removes it, `n % 2 == 0` means even.

Mixing types `7 / 2` is `3` but `7 / 2.0` is `3.5`. `static_cast<double>(a) / b` gives a real quotient. `int x = 3.9;` stores `3`.

Characters are numbers `'c' - 'a'` is `2`; `'7' - '0'` is `7`. `char col; char row; cin >> col >> row;` reads `e4` as `'e'` and `'4'`.

Comparison and logic `== != < <= > >=`; `&&` (and), `||` (or), `!` (not). A range is two tests: `x >= 1 && x <= 9`. `cout << (a < b);` prints `1` or `0`.

Selection

```cpp
if (x > 0) {
    cout << "positive";
} else if (x < 0) {
    cout << "negative";
} else {
    cout << "zero";
}
```

Absolute value and maximum without a library

```cpp
int d = a - b;
if (d < 0) { d = -d; }      // now d = |a - b|
int m = x;
if (y > m) { m = y; }       // now m = max(x, y)
```

Classic bugs `if (x = 5)` (use `==`); `if (x > 0);` (stray `;`); `int` overflow when a product exceeds 2.1×10^9 (use `long long`).

PART A — Arithmetic and types

60 points

These need only variables, `cin`/`cout` and arithmetic — **no `if` is required**.

A1 Convert the Temperature 5 pts

LeetCode 2469 · `leetcode.com/problems/convert-the-temperature`

Read a temperature in degrees Celsius (a real number, $0 \le c \le 1000$). Print it in Kelvin and in Fahrenheit:

$$\text{Kelvin} = c + 273.15 \qquad \text{Fahrenheit} = c \times 1.80 + 32.00$$

Input	Output	Why
`36.50`	`309.65 97.7`	
`122.11`	`395.26 251.798`	

TA: ____ / 5 pts initials ____

A2 Calculate Delayed Arrival Time 5 pts

LeetCode 2651 · `leetcode.com/problems/calculate-delayed-arrival-time`

A train is due at hour `arrivalTime` ($1 \le$ `arrivalTime` < 24) on a 24-hour clock and is `delayedTime` hours late ($1 \le$ `delayedTime` ≤ 24). Print the hour at which it actually arrives, from 0 to 23.

Input	Output	Why
`15 5`	`20`	15 + 5 = 20.
`13 11`	`0`	13 + 11 = 24, which is hour 0 of the next day.

TA: ____ / 5 pts initials ____

A3 Count of Matches in Tournament 5 pts

LeetCode 1688 · `leetcode.com/problems/count-of-matches-in-tournament`

A knockout tournament starts with n teams ($1 \le n \le 200$). In each round, teams are paired and each match eliminates the loser; if the number of teams is odd, one team gets a free pass to the next round. Rounds continue until one team is left. Print the total number of matches played.

Input	Output	Why
`7`	`6`	Rounds: 3 matches (7→4), 2 matches (4→2), 1 match (2→1).
`14`	`13`	7 + 3 + 2 + 1.

Hint: you do not need to simulate the rounds. How many teams must lose for one team to win?

TA: ____ / 5 pts initials ____

A4 Count Odd Numbers in an Interval Range 10 pts

LeetCode 1523 · `leetcode.com/problems/count-odd-numbers-in-an-interval-range`

Read two integers `low` and `high` ($0 \le$ `low` $\le$ `high` $\le 10^9$). Print how many odd numbers lie between them, **inclusive**. The range is up to a billion numbers wide, so counting them one by one is not an option — find a formula with `/`.

Input	Output	Why
`3 7`	`3`	The odd numbers are 3, 5, 7.
`8 10`	`1`	Only 9.

Hint: how many odd numbers are there from 1 to n? Try $n = 1, 2, 3, 4, 5$ on paper.

TA: ____ / 10 pts initials _____

A5 Determine Color of a Chessboard Square 10 pts

LeetCode 1812 · `leetcode.com/problems/determine-color-of-a-chessboard-square`

A chessboard square is named by a letter `a`–`h` (column) and a digit `1`–`8` (row); square `a1` is black and the colours alternate. Read a square such as `h3` and print `1` if it is **white**, `0` if it is black.

Input	Output	Why
`a1`	`0`	Black.
`h3`	`1`	White.
`c7`	`0`	Black.

Hint: read two `char`*s. Turn each into a number from 0 to 7 using* `'a'` *and* `'1'`*. What happens to the colour each time you move one square?*

TA: ____ / 10 pts initials _____

A6 Find the Maximum Achievable Number 5 pts

LeetCode 2769 · `leetcode.com/problems/find-the-maximum-achievable-number`

Read `num` and `t` ($1 \le$ each ≤ 50). A number x is *achievable* if, in at most t steps, you can make x equal to `num`, where one step adds 1 to (or subtracts 1 from) x **and at the same time** adds 1 to (or subtracts 1 from) `num`. Print the largest achievable x.

Input	Output	Why
`4 1`	`6`	One step: $x = 6 \to 5$ and $num = 4 \to 5$.
`3 2`	`7`	

Hint: in one step, by how much can the gap between x and `num` *shrink?*

TA: ____ / 5 pts initials _____

A7 Divisible and Non-divisible Sums Difference 10 pts

LeetCode 2894 · `leetcode.com/problems/divisible-and-non-divisible-sums-difference`

Read n and m ($1 \le n, m \le 1000$). Let `num1` be the sum of the numbers from 1 to n that are **not** divisible by m, and `num2` the sum of those that **are**. Print `num1 - num2`.

Input	Output	Why
`10 3`	`19`	$num2 = 3+6+9 = 18$, $num1 = 55-18 = 37$, $37-18 = 19$.
`5 6`	`15`	No multiples of 6: $15 - 0$.
`5 1`	`-15`	Every number is a multiple of 1: $0 - 15$.

Hint: $1 + 2 + \cdots + n = n(n+1)/2$. *How many multiples of m are there up to n, and what is their sum?*

TA: ____ / 10 pts initials _____

A8 Calculate Money in Leetcode Bank 10 pts

LeetCode 1716 · `leetcode.com/problems/calculate-money-in-leetcode-bank`

Hercy saves money every day. On the first Monday he puts in \$1, and each following day of the week \$1 more than the day before (Tuesday \$2, . . . , Sunday \$7). Every new Monday he puts in \$1 more than the previous Monday. Read the number of days n ($1 \le n \le 1000$) and print the total saved at the end of day n.

Input	Output	Why
`4`	`10`	$1 + 2 + 3 + 4$.
`10`	`37`	Week 1: $1 + \cdots + 7 = 28$; then $2 + 3 + 4 = 9$.
`20`	`96`	$28 + 35 + (3 + 4 + 5 + 6 + 7 + 8) = 96$.

Hint: `w = n / 7` *full weeks,* `d = n % 7` *extra days. Week 1 gives 28, and each later week gives 7 more than the one before.*

TA: ____ / 10 pts initials _____

PART B — Decisions with `if/else`

110 points

Use relational and logical operators and `if/else`. Test every branch, not just the examples.

B1 Smallest Even Multiple 5 pts

LeetCode 2413 · `leetcode.com/problems/smallest-even-multiple`

Read a positive integer n ($1 \le n \le 150$). Print the smallest positive number that is a multiple of both 2 and n.

Input	Output	Why
5	10	
6	6	6 is already even.

TA: ____ / 5 pts initials ____

B2 A Number After a Double Reversal 5 pts

LeetCode 2119 · `leetcode.com/problems/a-number-after-a-double-reversal`

Reversing a number means writing its digits backwards and dropping leading zeros: `2021` → `1202`, `12300` → `321`. Read `num` ($0 \le$ `num` $\le 10^6$), reverse it, then reverse the result. Print `true` if you get `num` back, otherwise `false`.

Input	Output	Why
526	true	526 → 625 → 526.
1800	false	1800 → 81 → 18.
0	true	

Hint: you do not have to reverse anything. When are digits lost?

TA: ____ / 5 pts initials ____

B3 Minimum Cuts to Divide a Circle 5 pts

LeetCode 2481 · `leetcode.com/problems/minimum-cuts-to-divide-a-circle`

A cut is a straight line through a circle: either a full diameter, or a radius from the centre to the edge. Read n ($1 \le n \le 100$) and print the minimum number of cuts that divide the circle into n equal slices.

Input	Output	Why
4	2	Two diameters.
3	3	Three radii; a diameter would make two of the slices unequal.

Hint: draw $n = 1, 2, 3, 4, 5, 6$. A diameter counts as one cut but makes two edges.

TA: ____ / 5 pts initials ____

B4 Harshad Number 10 pts

LeetCode 3099 · `leetcode.com/problems/harshad-number`

A number is a *Harshad number* if it is divisible by the sum of its digits. Read x ($1 \le x \le 100$). If x is a Harshad number, print the sum of its digits; otherwise print `-1`.

Input	Output	Why
18	9	1 + 8 = 9 and 18 mod 9 = 0.
23	-1	2 + 3 = 5, and 23 is not divisible by 5.

Hint: x has at most three digits. Take them apart with `/` and `%` (Lab 3, Worked example 5.1).

TA: ____ / 10 pts initials ____

B5 Type of Triangle 10 pts

LeetCode 3024 · leetcode.com/problems/type-of-triangle

Read three integer side lengths ($1 \le$ side ≤ 100). Print `none` if they cannot form a triangle (the sum of any two sides must be *greater* than the third); otherwise print `equilateral` (all equal), `isosceles` (exactly two equal) or `scalene` (all different).

Input	Output	Why
`3 3 3`	`equilateral`	
`3 4 5`	`scalene`	
`1 2 3`	`none`	1 + 2 = 3 is not greater than 3.

TA: ____ / 10 pts initials ____

B6 Categorize Box According to Criteria 10 pts

LeetCode 2525 · leetcode.com/problems/categorize-box-according-to-criteria

Read a box's `length`, `width`, `height` ($1 \le$ each $\le 10^5$) and `mass` ($1 \le$ `mass` ≤ 1000).

- The box is **bulky** if any dimension is at least 10^4, *or* its volume is at least 10^9.
- The box is **heavy** if its mass is at least 100.

Print `Both`, `Bulky`, `Heavy` or `Neither`.

Input	Output	Why
`1000 35 700 300`	`Heavy`	Volume 2.45×10^7 is small; mass 300.
`200 50 800 50`	`Neither`	

Warning: think about how big the volume can get before you choose its type.

TA: ____ / 10 pts initials ____

B7 K Items With the Maximum Sum 10 pts

LeetCode 2600 · leetcode.com/problems/k-items-with-the-maximum-sum

A bag holds `numOnes` items marked 1, `numZeros` items marked 0 and `numNegOnes` items marked -1 (each from 0 to 50). You pick exactly k items ($0 \le k \le$ total). Read the four numbers and print the largest possible sum.

Input	Output	Why
`3 2 0 2`	`2`	Take two 1s.
`3 2 0 4`	`3`	Three 1s and one 0.
`1 1 3 5`	`-2`	1 + 0 − 1 − 1 − 1.

TA: ____ / 10 pts initials ____

B8 Pass the Pillow 10 pts

LeetCode 2582 · `leetcode.com/problems/pass-the-pillow`

n people stand in a line, numbered 1 to n ($2 \le n \le 1000$). Person 1 holds a pillow. Every second the pillow is passed to the next person; when it reaches person n the direction reverses, and when it gets back to person 1 it reverses again. Read n and `time` ($1 \le$ `time` ≤ 1000) and print who holds the pillow after `time` seconds.

Input	Output	Why
`4 5`	`2`	$1 \to 2 \to 3 \to 4 \to 3 \to 2$.
`3 2`	`3`	$1 \to 2 \to 3$.

Hint: one full trip from one end to the other takes $n - 1$ seconds. Use `/` to count the finished trips and `%` for the seconds left over. What does the direction depend on?

TA: ____ / 10 pts initials _____

B9 Find Closest Person 5 pts

LeetCode 3516 · `leetcode.com/problems/find-closest-person`

Person 1 stands at position x, person 2 at y and person 3 at z on a number line ($1 \le x, y, z \le 100$). Persons 1 and 2 walk towards person 3 at the same speed. Print `1` if person 1 arrives first, `2` if person 2 does, and `0` if they arrive together.

Input	Output	Why
`2 7 4`	`1`	Distances 2 and 3.
`2 5 6`	`2`	Distances 4 and 1.
`1 5 3`	`0`	Both at distance 2.

TA: ____ / 5 pts initials _____

B10 Nim Game 5 pts

LeetCode 292 · `leetcode.com/problems/nim-game`

There are n stones on a table ($1 \le n \le 2^{31} - 1$). You and a friend take turns; you go first. On each turn a player removes 1, 2 or 3 stones. Whoever removes the last stone wins. Both players play perfectly. Print `true` if you can win, otherwise `false`.

Input	Output	Why
`4`	`false`	Whatever you take (1, 2 or 3), your friend takes the rest.
`1`	`true`	
`2`	`true`	

Hint: play it on paper for $n = 1$ to 8. What do the losing values have in common?

TA: ____ / 5 pts initials _____

B11 Rectangle Overlap 10 pts

LeetCode 836 · leetcode.com/problems/rectangle-overlap

A rectangle with sides parallel to the axes is given as `x1 y1 x2 y2`: its bottom-left corner $(x1, y1)$ and top-right corner $(x2, y2)$. Read two rectangles (8 integers, from -10^9 to 10^9) and print `true` if they overlap with a **positive area**, otherwise `false`. Touching only at an edge or a corner does not count.

Input	Output	Why
`0 0 2 2 1 1 3 3`	`true`	
`0 0 1 1 1 0 2 1`	`false`	They share only an edge.
`0 0 1 1 2 2 3 3`	`false`	

Hint: it is easier to describe when they do not *overlap: one is completely to the left of, right of, above or below the other.*

TA: ____ / 10 pts initials _____

B12 Angle Between Hands of a Clock 10 pts

LeetCode 1344 · leetcode.com/problems/angle-between-hands-of-a-clock

Read `hour` ($1 \le$ `hour` ≤ 12) and `minutes` ($0 \le$ `minutes` ≤ 59). Print the **smaller** angle, in degrees, between the hour hand and the minute hand.

Input	Output	Why
`12 30`	`165`	
`3 30`	`75`	
`3 15`	`7.5`	

Hint: the minute hand moves $6°$ *per minute. The hour hand moves* $30°$ *per hour* and $0.5°$ *per minute. Where is 12 o'clock?*

TA: ____ / 10 pts initials _____

B13 Total Distance Traveled 10 pts

LeetCode 2739 · leetcode.com/problems/total-distance-traveled

A truck has a main tank with `mainTank` litres and an additional tank with `additionalTank` litres ($1 \le$ each ≤ 100). It drives 10 km per litre, using only the main tank. Every time 5 litres have been used from the main tank, if the additional tank has at least 1 litre, 1 litre is moved into the main tank at once. Print the total distance the truck can drive.

Input	Output	Why
`5 10`	`60`	After 5 litres (50 km), 1 litre moves over: 10 km more.
`1 2`	`10`	Only 1 litre is ever used; no transfer happens.

Hint: each transfer needs 5 litres used but gives 1 back, so from the second transfer on it costs only 4 of the original litres. How many transfers can there be? Then there is a second limit: the additional tank.

TA: ____ / 10 pts initials _____

BONUS

+20 points

X1 Check if Two Chessboard Squares Have the Same Color +5 pts

LeetCode 3274 · `leetcode.com/problems/check-if-two-chessboard-squares-have-the-same-color`

Read two squares, for example `a1 c3`, and print `true` if they have the same colour, otherwise `false`.

Input	Output	Why
`a1 c3`	`true`	Both black.
`a1 h3`	`false`	

TA: ____ / 5 pts initials ____

X2 Determine if a Cell Is Reachable at a Given Time +5 pts

LeetCode 2849 · `leetcode.com/problems/determine-if-a-cell-is-reachable-at-a-given-time`

On an infinite grid you start at cell (sx, sy). Every second you **must** move to one of the 8 neighbouring cells (sideways or diagonally); cells may be revisited. Read `sx sy fx fy t` (coordinates from 1 to 10^9, $0 \le t \le 10^9$) and print `true` if you can be at (fx, fy) after *exactly* t seconds, otherwise `false`.

Input	Output	Why
`2 4 7 7 6`	`true`	The target is 5 columns and 3 rows away: 5 moves are enough, and 6 works too.
`3 1 7 3 3`	`false`	It is 4 columns away; 3 moves are not enough.

Hint: with diagonal moves, how many seconds does the shortest route take? Then think carefully about $t = 1$.

TA: ____ / 5 pts initials ____

X3 Moving Stones Until Consecutive +5 pts

LeetCode 1033 · `leetcode.com/problems/moving-stones-until-consecutive`

Three stones lie at different positions a, b, c on a number line ($1 \le$ each ≤ 100). In one move you pick up a stone at either end (the smallest or the largest position) and put it at an empty position *strictly between* the other two. The game ends when the stones are at three consecutive positions. Print the minimum and the maximum number of moves.

Input	Output	Why
`1 2 5`	`1 2`	Min: move 5 to 3. Max: 5 to 4, then 4 to 3.
`4 3 2`	`0 0`	Already consecutive.
`3 5 1`	`1 2`	

Hint: first sort the three numbers into $x < y < z$ using `if` and swaps.

TA: ____ / 5 pts initials ____

X4 Rectangle Area +5 pts

LeetCode 223 · `leetcode.com/problems/rectangle-area`

Read two axis-parallel rectangles as in B11 (`ax1 ay1 ax2 ay2 bx1 by1 bx2 by2`, each from -10^4 to 10^4) and print the total area they cover together — the overlap counted once.

Input	Output	Why
`-3 0 3 4 0 -1 9 2`	`45`	24 + 27 − 6.
`-2 -2 2 2 -2 -2 2 2`	`16`	The same rectangle twice.

TA: ____ / 5 pts initials _____

Score sheet

	Problem	Points	Earned	TA
A1	Convert the Temperature (LC 2469)	5		
A2	Calculate Delayed Arrival Time (LC 2651)	5		
A3	Count of Matches in Tournament (LC 1688)	5		
A4	Count Odd Numbers in an Interval Range (LC 1523)	10		
A5	Determine Color of a Chessboard Square (LC 1812)	10		
A6	Find the Maximum Achievable Number (LC 2769)	5		
A7	Divisible and Non-divisible Sums Diff. (LC 2894)	10		
A8	Calculate Money in Leetcode Bank (LC 1716)	10		
B1	Smallest Even Multiple (LC 2413)	5		
B2	A Number After a Double Reversal (LC 2119)	5		
B3	Minimum Cuts to Divide a Circle (LC 2481)	5		
B4	Harshad Number (LC 3099)	10		
B5	Type of Triangle (LC 3024)	10		
B6	Categorize Box According to Criteria (LC 2525)	10		
B7	K Items With the Maximum Sum (LC 2600)	10		
B8	Pass the Pillow (LC 2582)	10		
B9	Find Closest Person (LC 3516)	5		
B10	Nim Game (LC 292)	5		
B11	Rectangle Overlap (LC 836)	10		
B12	Angle Between Hands of a Clock (LC 1344)	10		
B13	Total Distance Traveled (LC 2739)	10		
X1	Same Colour Squares (LC 3274)	+5		
X2	Cell Reachable at a Given Time (LC 2849)	+5		
X3	Moving Stones Until Consecutive (LC 1033)	+5		
X4	Rectangle Area (LC 223)	+5		
	Sum (170 + 20 bonus available)	190		
	Score (capped at 100)	**100**		

After the lab

Every problem here is on LeetCode under the number shown. Once functions are covered (Week 6), rewrite your solutions as a LeetCode `class Solution` and submit them — the logic you wrote today stays exactly the same. Problems you did not finish today are good practice for Quiz 2 in Week 4.

End of Lab 4